PROJECT AND TEAM INFORMATION

Project Title

CRIMENET – Police Case and FIR Management System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Police case management involves handling FIRs, incidents, suspects, complainants, evidence and assigned officers. When these records are maintained separately, it can become difficult to quickly search cases, track their status and identify relationships between people or cases.
CRIMENET is proposed as a structured police case and FIR management system that provides authorized users with a unified environment for managing police case information. The system will support the registration, searching, updating and management of FIRs, incidents, persons, police officers and case records.
A distinctive feature of CRIMENET is its case connection analysis. The system will represent relationships between persons, cases and other relevant entities using graph-based data structures. Graph traversal can then be used to identify connections between suspects, persons and previously recorded cases.
The project is important because it demonstrates how Data Structures in C and Object-Oriented Programming in C++ can be combined to develop a practical case-management prototype.

State of the Art / Current solution

Modern police organizations generally use dedicated case-management and information systems for recording incidents, investigations, personnel and evidence. However, the exact software, workflows and access controls differ between organizations.
For an academic prototype, case information can be maintained using separate files or basic database tables. Such approaches can provide record storage and searching, but relationships between cases and persons may not be represented explicitly.
CRIMENET proposes a unified prototype in which FIR records, persons, officers, incidents and cases are connected through structured modules. Data Structures in C will support efficient organization and relationship analysis, while C++ will provide object-oriented modelling of system entities and user roles.

Project Goals and Milestones

Project Goals

- Develop a console-based police case and FIR management prototype.
- Provide role-based access for administrators, police officers and dispatchers.
- Store and manage FIR, incident, person, officer and case information.
- Implement searching, insertion, deletion and updating of records.
- Use Data Structures in C for efficient record organization and relationship analysis.
- Use C++ OOP concepts to model system entities and user roles.
- Implement graph-based case and suspect connection analysis.

Initial Milestones

Requirement analysis → system design → module development → data-structure integration → testing → final demonstration.

Project Approach

CRIMENET will be developed as a modular console-based application using both C and C++. The project will follow a role-based architecture containing three major user roles: **Administrator, Police Officer and Dispatcher**.

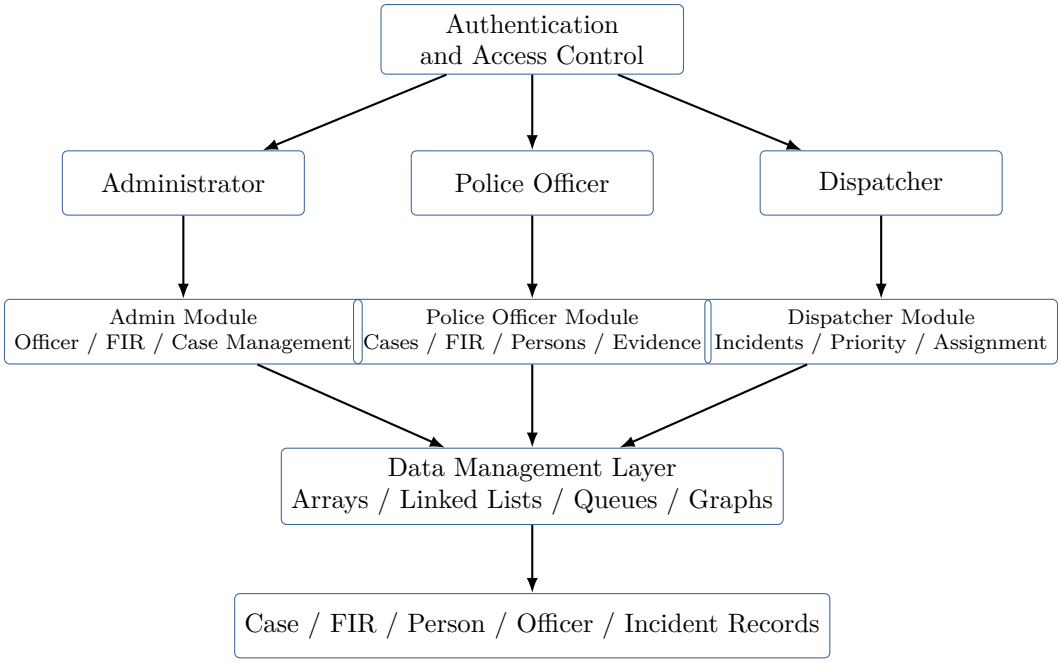
The Administrator will manage officers, dispatchers, FIRs, incidents, persons, case assignments and system records. The Dispatcher will receive incidents, maintain the priority queue, assign available officers and update incident status. The Police Officer will view assigned cases, register and update FIRs, manage persons and evidence, search records and perform case connection analysis.

C will be used for data-structure-oriented modules such as arrays, linked lists, queues and graphs. C++ will be used for classes, objects, inheritance and other OOP concepts.

The prototype will be developed using **Visual Studio Code** with **Git/GitHub** for version control and collaboration. The initial implementation will use structured file-based storage, with scope for database integration in later phases.

The development process will include requirement analysis, system design, implementation, module integration, testing and final demonstration.

System Architecture (High Level Diagram)



Major Components

- **Authentication Layer:** Provides role-based login for authorized users.
- **Admin Module:** Manages police officers, dispatchers, FIRs, incidents, persons and cases.
- **Dispatcher Module:** Receives incidents, manages priority and assigns officers.
- **Police Officer Module:** Handles assigned cases, FIRs, persons, evidence and investigations.
- **Data Structure Layer:** Uses arrays, linked lists, queues and graphs for efficient data organization.
- **Analysis Layer:** Identifies relationships and connections between cases and persons.

Project Outcome / Deliverables

The expected outcome is a functional console-based prototype named CRIMENET for police case and FIR management. The system will provide role-based interfaces for administrators, police officers and dispatchers. The major deliverables will include authentication and role management, FIR registration and management, incident management, officer assignment, person and suspect records, evidence management, case searching, investigation history and graph-based case connection analysis.
The project will also demonstrate meaningful integration of C Data Structures and C++ Object-Oriented Programming concepts. A modular source-code structure, system architecture, test cases and project documentation will be prepared as part of the deliverables.

Assumptions

The project is assumed to be an academic prototype and not a production police information system. Only authorized users are assumed to access the system. Sample or synthetic records will be used for testing. The prototype will initially use file-based storage and structured in-memory data structures. Real-world requirements such as biometric authentication, live GPS tracking, advanced security infrastructure and direct integration with government databases are outside the initial scope.

References

1. Bjarne Stroustrup, *The C++ Programming Language*.
2. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*.
3. Thomas H. Cormen et al., *Introduction to Algorithms*.
4. Official C++ documentation and standard library references.
5. Course material for Data Structures in C and Object-Oriented Programming with C++.